

What are the chances of an asteroid hitting you?

你被一颗小行星撞击的概率有多大？

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科学家们直到二十世纪晚期才发现，其实许多小行星的运行轨道经常会接近地球。虽然这些小行星很可能只是与我们“擦肩而过”，但发生撞击的概率并非为零。那么，我们被一颗小行星撞击的概率有多大？最近，一颗代号为 2025 YR4 的小行星受到了天文学家的持续追踪，因为它被预测有 2.3% 的概率在 2032 年撞击地球。本文浅谈小行星撞击地球的概率有多大。

词汇: asteroids 小行星

- 1 An asteroid ended the age of the dinosaurs 66 million years ago. Could something as catastrophic as that happen again?
- 2 Asteroids are something astronomers closely track. Recently, the asteroid named 2024 YR4 made headlines for its tiny, but real, chance of colliding with Earth in 2032. In February 2025, NASA increased the impact probability of this asteroid hitting us to 2.3%. Scientists have calculated that YR4 is between 40m and 90m in diameter. It would have the power of a nuclear bomb if it was to hit Earth, causing severe damage if the impact was in a populated area. However, Dr Robert Massey of the Royal Astronomical Society urges calm. He said he is not "losing sleep over" the asteroid, but stresses that funding planetary defence and spotting potential threats is essential.
- 3 So what are asteroids? They are fragments of rock left over from the creation of the solar system 4.6 billion years ago. They frequently orbit close to Earth pushed by the gravity of other planets, though it was only in the late 1900s that scientists were able to systematically track near-Earth objects and learn this. They found that asteroids of more than 40m across, that's large enough to destroy a city, cross the Earth's path within lunar distance several times a year.
- 4 The chance of an asteroid being large enough to penetrate our atmosphere intact and destroy a city is likely to happen once in every hundred years. In 1908, an asteroid exploded over Siberia, injuring people and buildings across more than 500 square kilometres. However, actual impacts aren't as common as near-misses. For example, the asteroid Apophis raised alarms after it was discovered in 2004 because it was estimated to be around the size of a cruise ship. Even scarier, in 1994, the comet named Shoemaker-Levy 9 crashed into Jupiter, and, if it had hit Earth, all life would have been wiped out, just like the dinosaurs were.

5 So, it's safe to enjoy admiring the night sky because the chances of you being hit by an asteroid are very, very small, even if they're not zero.

测验与练习

一、为本文选择总结句。

a. Asteroids are more common than you think. b. Asteroids are unlikely to impact you. c. Asteroids are more of a danger than ever.

二、根据文章内容选出正确的选项。

1. Scientists expect 2024 YR4 to hit Earth in the next ten years. a. True b. False c. Not given
2. What does Dr Massey say is very important? a. remaining calm in the event of an impact b. ensuring budget is spent on space trips c. continuing to research the likelihood of asteroid impacts
3. Asteroids come close to our planet because of... a. the gravitational pull of Earth. b. lunar forces impacting Earth. c. the gravitational push from other planets.
4. What does 'this' refer to in the following sentence? ... it was only in the late 1900s that scientists were able to systematically track near-Earth objects and learn this . a. That asteroids regularly come close to Earth. b. Gravity plays a role in the movement of asteroids. c. That asteroids of 40m in diameter could destroy a city.
5. Apophis was a large asteroid that hit Earth in 2004. a. True b. False c. Not given

三、用下列单词完成对本文的总结。

A/an 1) _____ could 2) _____ humanity, but it's unlikely. 3) _____ are more common than impacts, though 4) _____ is something that money should be spent on because the chance isn't zero. An asteroid the size of YR4 could hit with the power of a 5) _____, for example.

planetary defence near-misses nuclear bomb fragments wipe out asteroid intact

(答案见词汇表后)

词汇表

astronomer

天文学家 collide 碰撞 diameter 直径 nuclear bomb 核弹 lose sleep 失眠 planetary defence 行星防卫
fragment 碎片 left over 被剩下的 solar system 太阳系 orbit 沿轨道运行 gravity 重力 systematically 系统性
地 penetrate 穿透, 突破 intact 完好无损的 near-miss 擦肩而过, 差点击中 raise alarm 发出警报, 引起警惕
wipe out 彻底摧毁

答案

一、为本文选择总结句。

b. The chances of you being hit are very, very small , but they're not zero.

二、根据文章内容选出正确的选项。

1. b. False. In February 2025, NASA increased the impact probability of this asteroid hitting us to 2.3%.

2. c. ...but stresses that funding planetary defence and spotting potential threats is essential.

3. c. They frequently orbit close to Earth pushed by the gravity of other planets .

4. a. They frequently orbit close to Earth pushed by the gravity of other planets, though it was only in the late 1900s that scientists were able to systematically track near-Earth objects and learn this .

5. c. Not given . It says the asteroid raised alarms, meaning caused people to worry, but doesn't say if it hit Earth or not. The asteroid Apophis raised alarms after it was discovered in 2004 because it was estimated to be around the size of a cruise ship.

三、用下列单词完成对本文的总结。

An asteroid could wipe out humanity, but it's unlikely. Near-misses are more common than impacts, though planetary defence is something that money should be spent on because the chance isn't zero. An asteroid the size of YR4 could hit with the power of a nuclear bomb , for example.

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